

Subhan Sajadi

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Education

BEng, Mechanical Engineering (Mechatronics Option), Minor in Computer Science Sep 2022 – Apr 2027
Toronto Metropolitan University • Ontario Scholar • Entrance Scholarship
Coursework: Robotics, Sensors & Instrumentation, Manufacturing System Control, Vibrations, Microprocessor Systems

Research & Professional Experience

SymBioRobotX Lab (Wearable Exosuits), Toronto Metropolitan University Toronto, ON
Undergraduate Research Assistant Jul 2026 – Present

- Built a **MATLAB/Simulink Desktop Real-Time** controller for a CubeMars AK70-9 exosuit actuator over **CAN** with position, velocity, and current modes, a safe-disable path, and live feedback; kernel mode removed a ~ 4 s lag.
- Traced silent driver clamping to a $10,000\times$ scaling error and a wrong current-loop ID; measured $\sim 80\%$ tracking at 1 Hz.
- Designed a 12-pin CAN/48 V tether with bus termination and a pre-connection safety test, cutting the BOM to $\sim$ US\$233.

Miniaturization in Bioanalysis Lab, iBEST (St. Michael's Hospital) Toronto, ON
Undergraduate Research Assistant May 2026 – Aug 2026

- Developed a two-layer **SU-8 photolithography** process on 100 mm wafers (MicroWriter ML3 maskless lithography) to fabricate mother-machine microfluidic devices for single-cell *E. coli* imaging.
- Diagnosed resist-thickness discrepancies via optical profilometry; authored the lab's 30-section cleanroom SOP (v3.0).

Litens Automotive Group Vaughan, ON
Purchasing Student May 2025 – May 2026

- Built an **Excel/Power Query** cost-benchmarking tool behind projected multi-million-dollar savings.
- Reviewed quotes against drawings, BOMs, and machining, forging, casting, and moulding cost drivers, finding design changes worth up to 10%; automated PO workflows in **VBA**, saving 2 h/day with zero errors.

Celestica Toronto, ON
Commodity Management May 2024 – Aug 2024

- Analyzed BOMs and price databases for competitive global bids; negotiated supplier contracts for 10% added savings.

Technical Projects & Design Teams

Computer Vision Developer, TMU Wave Autonomous Underwater Vehicle Team Mar 2026 – Present

- Developing **OpenCV** gate, slalom, and path detectors for an AUV in a Dockerized ArduPilot SITL/MAVLink stack.

1/10-Scale Autonomous Race Car, TMU Velocity (F1Tenth)

- Implemented SLAM-based localization with RViz visualization in **ROS2**, reaching sub-100 ms perception-to-control latency.
- Engineered LiDAR emergency braking with time-to-collision logic, cutting collision events by $\sim 90\%$ in testing.

Smart Rehabilitation Knee Sleeve, True North Biomedical Competition Oct 2025 – Mar 2026

- Designed and prototyped a knee sleeve fusing surface **EMG** and **IMU** sensing to quantify muscle-activation asymmetry and knee range of motion for ACL rehab; modelled the CAD assembly and BOM and completed a Class I regulatory analysis.

Rocker-Bogie Competition Rover, Take-Off Robotics Competition Winter 2026

- Built a 6-wheel rocker-bogie rover with a 3-DOF servo arm and claw, adapting an open-source suspension into a custom **SolidWorks** chassis; programmed **ESP32** PWM drive and servo control.

Leadership

VP Competitions, Ontario Engineering Competition 2027 Jan 2026 – Present

- Co-lead 9 provincial competitions for 300+ students, directly overseeing 6 with ~ 20 leads and a \$5K build budget.
- Built OEC Tracker (**React**, **Supabase**, **Node**): LLM-extracted action items with human review and Discord reminders.

VP Events, Mechatronics Course Union May 2025 – Present

- Lead 4 directors and 6 first-year reps running 6+ workshops for 150+ students; built the React/Supabase events dashboard.

VP Competitions, Metropolitan Engineering Competition May 2023 – Nov 2025

- Ran 18 competitions with 26 coordinators and $\sim$ \$4K procurement; helped build an inventory website with ~ 500 active users.

Skills

CAD & Analysis: SolidWorks, Fusion 360, Onshape, AutoCAD, ANSYS FEA, GD&T, Engineering Drawings, MATLAB

Embedded & Controls: ESP32, Arduino, CAN Bus (PCAN), I2C/SPI, Simulink Desktop Real-Time, KiCad, FPGA/VHDL

Software & Robotics: Python, C/C++, Java, TypeScript/React, SQL, VBA, Git, Docker, ROS2, RViz, OpenCV, YOLOv8

Fabrication & Lab: 3D Printing (FDM), CNC Machining, CMM & Metrology, SU-8 Photolithography, PDMS Soft Lithography

Certifications: Controlled Goods Program, WHMIS